Association for Project Management

Budgeting and cost control

Learning objective:

22. Understand budgeting and cost control as the ability to estimate costs, develop and agree budgets and monitor actual costs against forecast costs.

APM-QS-PMQ2-24a V1

22a) Know how to create a budget (including the use of a cost breakdown structure) and the different costs included in a budget (fixed, variable, direct, indirect etc.)

Creating a budget

In the project profession, estimates are used for multiple purposes, one of which is to support budgeting and cost control. Budgets are created from estimates, looking at the approximation of time and cost to complete the scope of work to the required acceptance criteria.

Estimates are used to:

- Enable budget setting and considerations of affordability.
- Support judgements about value for money for the solution chosen.
- Create a resource schedule which then in turn provides the cost view of the project.

As part of creating a budget, consideration should be given to:

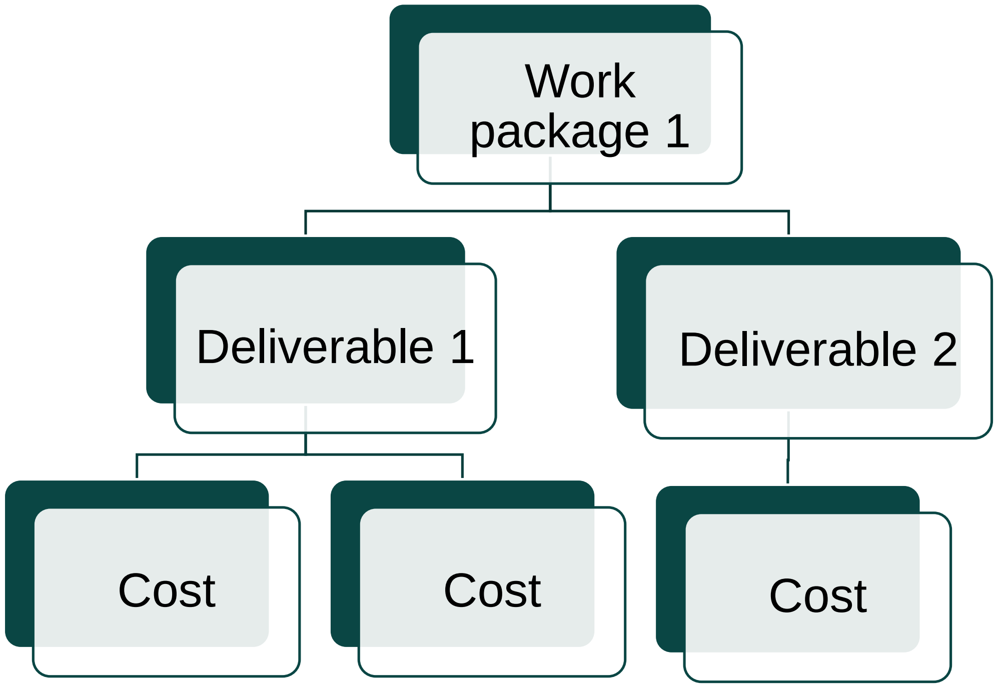
- The different types of costs involved and when they will need to be paid so that they can be budgeted accordingly, i.e. funding drawdowns to cover payment milestones.
- Information required from external sources, such as supplier costs, and how these align to the plan to enable budgeting.
- The assumptions being made on the project and how they underpin the cost estimates so they can be understood, validated and budgeted for.

The cost breakdown structure (CBS)

- The work breakdown structure (WBS) is used to create the CBS by taking each of the work packages defined for the project and looking at the estimated costs related to the delivery of activity. Through this process, a budget can be set for the overall project.

- The CBS identifies all activity that has a cost associated with it, both labour and non-labour, in each work package. These costs can then be collected, recorded, monitored and controlled as part of the financial reporting system. Costs can be attributed to:
    - people
    - equipment
    - materials
    - other resources required

- The CBS will reflect the financial coding used in the project for booking and recording. This may be set at an organisational level as part of standard financial reporting procedures.



The relationship between different costs

- Knowing the overall cost of the project is a good start; however, on its own this is not enough for project control.
- It is key to know, within that overall cost, where the costs fall over time in the schedule.
- It is then possible to manage resources, supplier payments and funding requests effectively.
- The project life cycle determines the frequency with which funds are to be released.
    - In a linear life cycle, funds may only be released at decision gates when the costs spent to date are understood and future forecasts have been approved.
    - In an iterative life cycle, the release of funds may be more frequent due to work being completed in short intervals.
- This cost planning exercise is an essential part of the process to manage cashflow and to ensure the project manager knows when funding drawdowns are needed.

The relationship between different costs

Cost planning aspects include looking at the type of costs:

- **Are costs fixed?** For example, a contract may set a monthly cost for resources provided by a particular supplier, irrespective of the actual resources used by the supplier over a month.
- **Or are they variable?** For example, paying for materials as and when they are needed or used. The monthly cost may well be different from month to month.
- **Are costs recurring?** For example, software licenses and maintenance reoccur at set periods, happen periodically as an event happens and contribute multiple costs to the project at different times.
- **Or are they non-recurring?** For example, equipment set-up costs are a one-off cost to the project, happening once in a project life and contributing a single cost.

It's important to consider whether costs are direct or indirect, as this makes a difference to how easy it is to identify, track and manage them. Direct costs are directly related to the outputs of the project, whereas indirect costs are associated with the costs of operating the outputs for the business as a whole.

22b) Know how to forecast and refine budgets using cost control techniques e.g. earned value.

Cost control techniques

Cost control is necessary to identify variances to project budgets and cost forecasts to inform decision making.

Project managers use cost control techniques to:

- Minimise costs where possible.
- Identify areas of overspend.
- Correct unacceptable levels of overspend.
- Inform future projects by providing insight for lessons learnt.

Forecasts are created and used to determine project costs against scheduled activities.

When changes are made to the agreed baseline, it is important to update the cost information to refine budgets.

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Earned value management

A recognised best practice approach to cost control is earned value management (EVM), which is ideal for identifying where a project is not progressing efficiently.

- Earned value analysis reports on the project spend against the spend curve and the value of the work achieved. If the project is underspent or overspent against the planned spend line, knowing the value of work done at that point helps to refine budgets based on current project status.
- The efficiency of spend is looked at through the cost performance indicator (CPI) and productivity through the schedule performance indicator (SPI).
- A pictorial representation of project progress can be provided by tracking the earned value on an earned value graph.

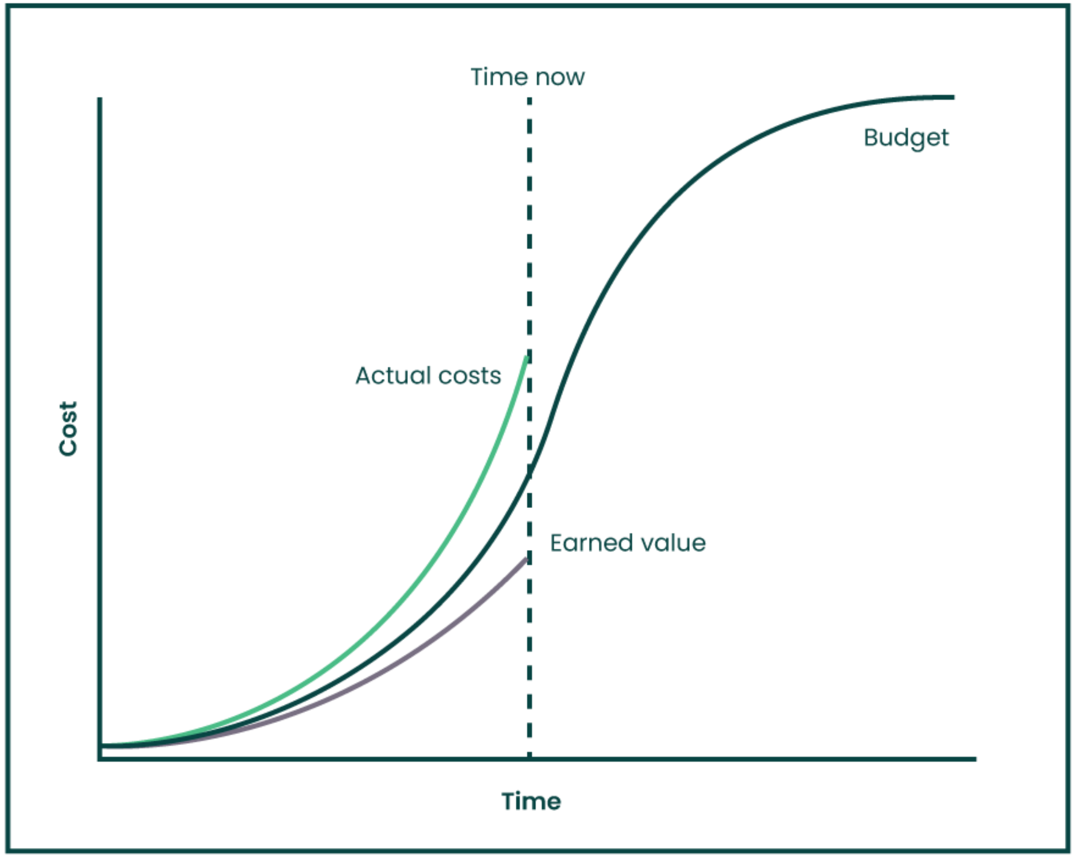
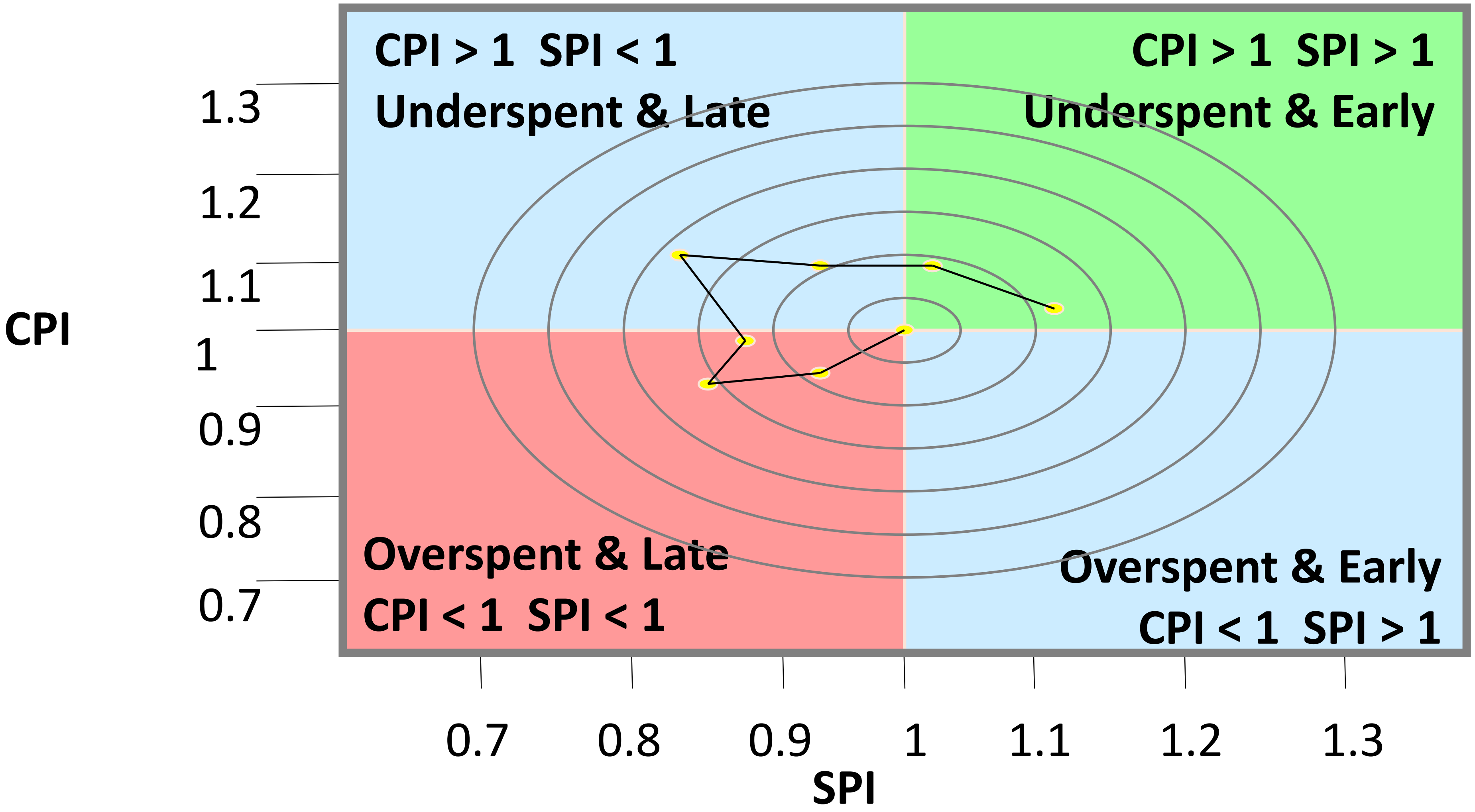


Figure 2: Budget vs actual plus earned value graph
Taken from the Earned Value Management: APM Guidelines (2008)

Earned Value (EV) equates to the value of the work ACTUALLY PERFORMED, and EVM compares this to planned work and the actual cost charged

EVM Monitoring



22c) Know how to monitor and report financial performance (including different types of financial reports).

Monitoring and reporting financial performance

Monitoring takes the baseline that has been agreed and looks at how the project is performing against it. It enables meaningful reports to be shared to improve performance and enable decision making.

- There are different approaches that can be taken to progress monitoring and reporting. The project manager and sponsor agree appropriate methods to monitor financial performance early in the project.
- Monthly monitoring is often appropriate.
- A traffic light approach can be used when reporting to flag the areas that are in or out of the agreed forecast including tolerance levels.
- Financial performance is monitored through different reviews that happen as part of project governance. The following all monitor progress against baselined budgets, timelines and success criteria:
  - decision gates
  - benefit reviews
  - stage reviews
  - audits

Monitoring and reporting financial performance

Usual reporting could include:

- actual costs versus forecasted costs
- actual spend versus forecasted spend against a set spend curve
- actual spend versus actual work achieved looking at the efficiency of spend (earned value analysis)
- estimated project budget and the forecasted cost overrun based on planned completion dates
- cashflow and drawdown requests
- resource costs
- change request costs
- contingency drawdown requests
- financial benefit realisation

22d) Know how to close down finances at the end of a project.

Closing down finances at the end of a project

When the project has completed delivery, as part of governance, the project sponsor will commence close down activities. Typically, this will involve a project closure report, which is then submitted to the final decision gate for endorsement. As well as the report, the following activities need to take place:

Reassign resources

- Move resources back into the business, back to the supplier or out of the organisation.
- Ensure organisation is not charged for resources that are no longer required.

Formally close contracts

- Facilitate closing all contracts in the supply chain.
- Make sure everything has been done, and everything expected to be received has been.
- Agree final accounts.

Accounting and management systems updated

- Close down project on the accounting and management systems.
- Remove delegated authorities and project cost codes.
- No further money can be charged against the project.

Surplus funding returned

- Once all payments have been settled, any budget not spent returns to the business for use elsewhere.

Consider how to close down activities early in the project life cycle so as not to incur any costs that have not been factored into the budget.